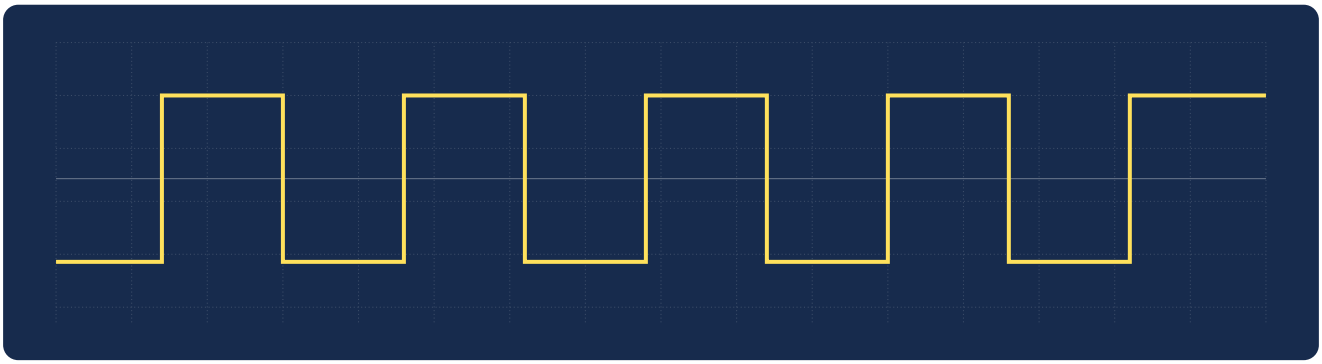
DSO138
Oscilloscope

User manual

Classic board | C++ firmware



4 keys
Run, hold and edit

9 ranges
10 mV/div to 5 V/div

18 timebases
10 μs/div to 5 s/div

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Manual revision 2.0 | 3 November 2026

Firmware 2.0.0 source (UID-configured build required)

Intended hardware: PCB 109-13800-00C, STM32F103C8T6, ILI9341 display.

This guide describes the implemented controls and functions. Nominal readings are not a calibration certificate.

1 Quick start

> **Board and power**
>
> This guide is for the identified DSO138 board, PCB 109-13800-00C, with STM32F103C8T6 and ILI9341 display. Use the board's normal 9 V DC input as marked on the board. For bench work, use a low-voltage isolated supply and connect signal ground to the scope ground. Use only isolated low-voltage bench signals; never connect the scope input or ground directly to mains. Do not infer a safe maximum input from this guide; observe the board markings and its analog input limits.

This operating guide assumes a correctly configured firmware image. Set your MCU UID and board identity before building and programming. See the repository's Getting started guide, including the upgrade notes for settings saved by earlier releases.

Try the built-in signal

1. Connect the onboard test output to the signal input.
2. Set the physical coupling switch to DC, base sensitivity to 1 V/div, and multiplier to $\times 1$.
3. Press **SEL** until Timebase is selected. Use **+** / **-** to select 1 ms/div.
4. Select Trigger mode and choose AUTO with **+** / **-**.
5. In an ordinary field such as Timebase, press **OK** if the scope is in HOLD. Look for RUN and a repeating square wave.
6. Press **OK** again to hold the record for inspection.

The observed board showed about 3.3 vertical divisions and a frequency near 1 kHz in this setup. This is a functional check, not a guaranteed voltage, frequency standard, or calibration reference.

On power-up, saved valid settings return in HOLD; press **OK** to run. If there are no saved settings, the enabled board profile requests Auto run after the physical switches settle. Waveform save and recall are not available. Allow the display to show a record before judging a signal.

The three slide switches set the analog input. Their positions physically select coupling and sensitivity. Software settings cannot move them. After changing a switch, wait for the on-screen status to leave `WAIT INPUT` or `Switches settling` before using a new record.

2 Buttons and physical switches

Control	Action
SEL	Move to the next setting field. After Domain, the sequence returns to Timebase.
+ , -	Change a value or cycle a choice. For numeric settings, holding a button repeats after about 400 ms, then at about 80 ms intervals. Discrete choices do not repeat.
OK	Run or hold in ordinary fields; rearm Single when appropriate. In Storage or Calibration, the first press stops acquisition and a later press executes the selected action. Release between presses.
RESET	Hardware reset. Unsaved edits are lost; the last saved configuration normally returns in HOLD. A reset during flash erase/program can lose saved data; see Saving.

The controls accept one button action at a time. If buttons are pressed together, release them before trying again. A completed Single capture or a fault may need an acknowledgement before the next action; wait for the status to settle, release, then press OK again when the screen indicates the action is ready.

Base sensitivity	Multiplier $\times 1$	Multiplier $\times 2$	Multiplier $\times 5$
10 mV/div	10 mV/div	20 mV/div	50 mV/div
100 mV/div	100 mV/div	200 mV/div	500 mV/div
1 V/div	1 V/div	2 V/div	5 V/div

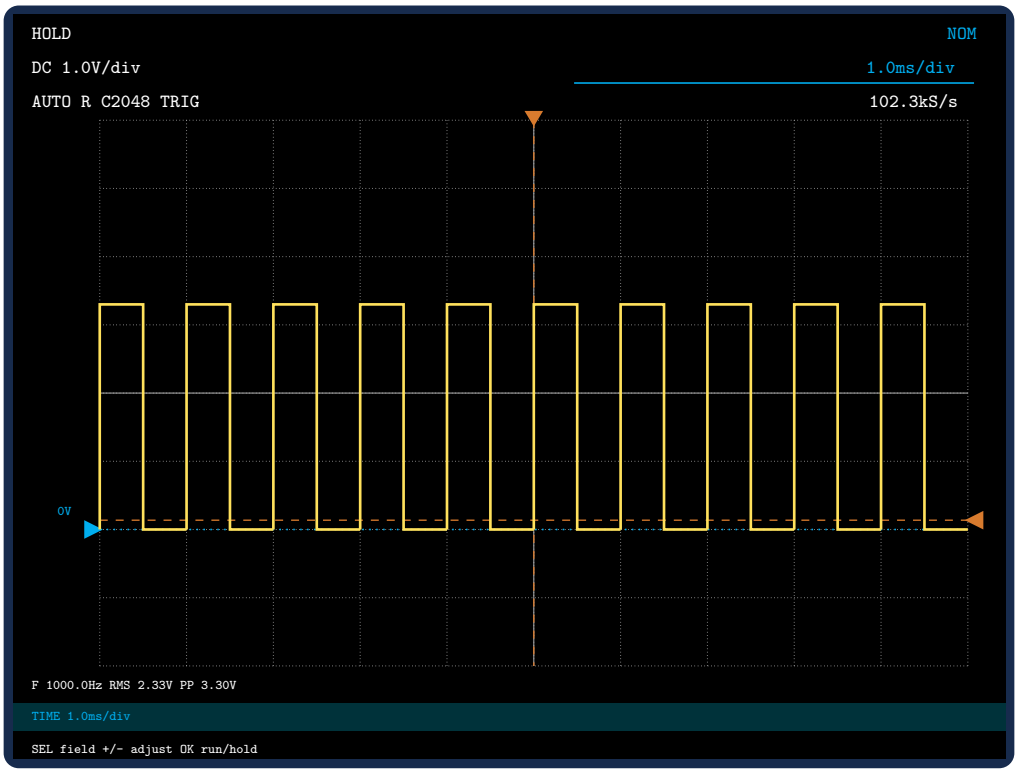
The separate coupling switch selects DC, AC, or GND.

The table gives the nine nominal ranges. The separate coupling switch selects DC (shows the input offset), AC (blocks DC), or GND (selects the internal ground path). These are front-panel sensitivity labels. They do not confirm the input voltage or a probe's attenuation. If using a $\times 10$ probe, account for that factor yourself; there is no probe-attenuation menu.

Trigger activity LED

D3 gives a brief, about 100 ms indication for an accepted trigger crossing; another crossing restarts the pulse, so close events may look continuously lit. Auto free-run timeout and HOLD redraw do not light it. A Single pulse can finish through automatic HOLD; faults leave it off. The indicator has been observed working on the reference instrument; its pulse timing has not been instrument-measured, so it is not a precise timing strobe.

3 Reading the screen



Illustrative layout, not a screenshot or a measured record.

The display status tells you what the instrument is doing: `RUN` or `ARM` indicates active acquisition or arming, `HOLD` indicates a stopped view, and `WAIT INPUT` or `WAIT BACK` means it is waiting for stable switches or an acquisition backend. The selected Domain controls whether the display shows the scope trace or the spectrum of a completed record.

The plot has 10 horizontal and 8 vertical divisions. The yellow trace/envelope shows available samples; drawing does not create extra samples. Orange marks the trigger position and level; cyan marks the zero reference and selected control. Focus labels beginning `NEXT` show the value to be used by the next capture, while a held trace continues to show its frozen capture settings. The reference line is labeled `0V` with a usable voltage profile, or `REF` when values are raw ADC codes.

The top labels show coupling, sensitivity, timebase, trigger mode/edge/level and sample spacing or rate. A held waveform keeps the range, coupling, timing and calibration provenance from its capture. If a slide switch changes, the old held trace is not rescaled to the new analog setting; the display may say `OLD range/coupling held`. Start a fresh capture to inspect the new setting.

Reading the grid: 3.3 vertical divisions at 1 V/div represent about 3.3 V peak-to-peak. One full cycle spanning one horizontal division at 1 ms/div represents a 1 ms period, or about 1 kHz. Apply an external probe's attenuation factor separately.

The quality tag describes voltage conversion: `NOM` uses nominal board coefficients, `CAL` means a measured profile is installed, and `RAW` means no voltage conversion is available. None of these labels by itself certifies measurement accuracy. Clipping warnings mean amplitude and frequency results are invalid for that record.

4 Menu and settings

Press SEL to move through the thirteen fields. Use + and - to adjust; use OK as shown. The focused field is shown near the bottom of the screen.

Focus	+ / -	OK
Timebase	Select a catalogue setting	Run/Hold
Trigger mode	AUTO, NORMAL, SINGLE	Run/Hold; Single rearm
Trigger edge	Rising or falling	Run/Hold; Single rearm
Trigger level	ADC threshold, 16-code steps	Run/Hold; Single rearm
Trigger position	0 to 100%, 2.5% steps	Run/Hold; Single rearm
Vertical position	Move reference, 2-pixel steps	Run/Hold; Single rearm
Horizontal position	Pan, 16-sample steps	Run/Hold; Single rearm
Measurements	Overview, Amplitude, RMS/Mean, Freq	Run/Hold
Storage	Choose one of four actions	Stop, then execute
Calibration reference	Set signed reference value	Stop, then cycle step
Calibration tolerance	Set allowed error	Stop, then cycle step
Calibration	Begin, Accept Ref, Install, Abort	Stop, then execute
Domain	Spectrum or Scope	Run/Hold

For Storage and Calibration, if acquisition is running, the first OK changes to HOLD. Release the button, then press OK again to carry out the selected action. In Reference and Tolerance, OK in HOLD cycles the decimal edit step; + and - change the value by that step. In Domain, + selects Spectrum and - selects Scope; the choice is session-only and the instrument starts in Scope after boot. `SAVE CONFIG` still saves the eight scope settings listed under Storage; it does not save the Domain choice. Reference/tolerance editors are not saved by that action.

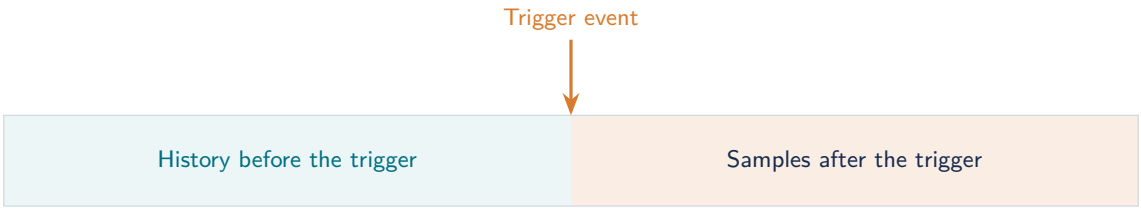
5 Triggering and viewing

- **AUTO:** publishes a record when a trigger edge qualifies, or after its timeout if no qualifying edge arrives. Use it for a general live view and for the built-in signal check.
- **NORMAL:** waits for the selected edge at the trigger level. If the signal never crosses the level in the selected direction, it can keep waiting.
- **SINGLE:** captures one complete record and stops. Press OK to arm another capture. If the completed record or a fault has not yet been acknowledged, wait for the screen state, release, and press again.

Choose rising or falling edge in Trigger edge. Trigger level changes in 16 ADC-code steps; it is a code threshold, not a calibrated voltage setting. Trigger position moves from 0% to 100% of the record in 2.5% steps, selecting how much history lies before the trigger in ordinary triggered captures.

Timebase, trigger mode, edge, level and trigger position affect capture. Changing them during a run invalidates the incomplete record. On a held trace, timebase changes zoom the existing samples; they do not resample the waveform. Areas outside the captured record remain empty. Vertical + moves the reference down; - moves it up. Horizontal + pans toward later source samples; - pans toward earlier samples, in 16-sample steps. These view adjustments do not make a new capture.

Example: trigger position at 50%



Record position before any view pan or timebase zoom.

Use HOLD to inspect a stable record, then RUN to resume. With AC coupling, the instrument cannot show the DC component removed by the input circuit. With GND coupling, the trace shows the grounded input path, not an external signal.

6 Measurements and validity

Measurements cover the complete finite record, not just the visible portion or a hand-selected cycle. The four measurement pages are Overview, Amplitude, RMS/Mean and Freq. Amplitude reports peak-to-peak only; RMS/Mean reports total RMS and the arithmetic mean. The scope measurement-focus line also shows absolute peak magnitude when voltage conversion is available.

Screen item	Meaning
PP	Peak-to-peak voltage on the Amplitude page.
PK	Absolute peak voltage from the conversion zero, shown on the scope measurement-focus line. With a DC offset, it is not necessarily half of PP.
RMS	Total RMS about the calibration zero, including DC, on the RMS/Mean page.
MEAN	Arithmetic mean on the RMS/Mean page.
F	Detected repetition rate, when valid.

With `RAW`, the display reports ADC counts (marked c) rather than input volts. `NOM` uses the board's nominal conversion coefficients. `CAL` means an installed measured DC profile is being used. A displayed voltage is a converted estimate, not an accuracy guarantee. AC-coupled mean and total RMS describe the AC-coupled signal; they cannot restore removed DC.

Frequency needs at least three qualified complete periods. A clear trace can still show `-- CYCLES` when it is too short. Other reasons can include low amplitude, instability, clipping, grounding or a rate outside estimator limits. Frequency is observed repetition, not proof of the signal's fundamental frequency. Aliasing can produce a plausible but incorrect value.

An ADC rail hit suppresses valid amplitude and frequency claims; the scope view may show `ADC CLIP`, `RAW ADC CLIP`, or `F --CLIP; volts invalid`. A missing voltage profile gives `RAW: no volts`. Stop and correct the input/range before relying on another record. Clipping can add harmonics to the Spectrum view; it has no separate clipping warning, so check the same record in Scope view. As a mathematical illustration only, an ideal 0-to-3.3 V, 50% duty square wave has 3.3 V PP, 1.65 V mean and 2.33 V total RMS. These calculated values are not a test or specification of the onboard output.

7 Spectrum view

Spectrum view analyzes 512 consecutive samples from a completed, frozen record. The calculation uses the record's known timer rate and the same sample block throughout. With RUN active, a new completed capture produces a refreshed spectrum; analysis runs after capture, not while the ADC is sampling. HOLD records can also be viewed. A record must contain at least 512 samples and have known timing.

The analyzer subtracts the block mean and applies a fixed periodic Hann window before the 512-point FFT. Mean removal suppresses DC, so a constant or zero input has no positive-frequency peak and the readout shows `Peak --`. The Hann window reduces leakage from tones between frequency bins, while spreading a tone over nearby bins. The peak readout uses the strongest positive-frequency bin without interpolation, so its frequency is quantized in steps of $\Delta f = F_s/512$.

Display item	Meaning
Horizontal axis	From 0 Hz to the Nyquist frequency, $F_s/2$. The label `df` gives the bin spacing $F_s/512$. F_s is the configured sample rate derived from the timer plan, not an independently measured clock or accuracy specification.
Vertical axis	Fixed 80 dB span from +10 to -70 dBFS, at 10 dB/div. The 0 dBFS marker is 22 pixels below the plot top.
`Pk` readout	Strongest positive-frequency bin, in hertz, followed by its level in dBFS.

dBFS uses a peak reference of 2048 ADC codes: 0 dBFS corresponds to a 2048-code sinusoidal peak in the input-code scale. It is not a voltage reading, and the Spectrum view does not use the installed voltage calibration. FFT length, window and vertical scale are fixed. The horizontal axis always shows the full Nyquist span for the captured sample rate; changing the acquisition timebase can change that span.

Nyquist is the highest frequency represented without folding for the sampled block. Higher-frequency energy can alias into the displayed band and appear as a plausible lower-frequency peak; the Hann window does not prevent aliasing. Use suitable analog bandwidth limits ahead of the ADC when the signal may contain energy above Nyquist. ADC clipping can create harmonics that look like real signal components. Spectrum view has no clipping warning; check the same record in Scope view for its ADC clipping status.

Accessing the spectrum

Use **SEL** to focus Domain, then press + for Spectrum or - for Scope. In RUN, each new complete record is analyzed after capture; in HOLD, the current record remains available. The Domain choice is not stored by `SAVE CONFIG` and resets to Scope after restart.

8 Settings and calibration storage

Storage has four actions. Select one with + / -. If acquisition is running, the first OK changes to HOLD; release the button, then press OK again to execute. Wait for SAVED after a save before changing power or resetting.

Displayed action	What it does
SAVE CONFIG	Save the eight scope settings: timebase, trigger mode/edge/level/position, vertical/horizontal position and measurement page. Domain is session-only.
LOAD CONFIG	Restore saved scope settings. The restored settings start in HOLD.
SAVE CAL	Persist the currently installed DC calibration profile.
RESTORE CAL	Restore the saved DC calibration profile into the live profile. A valid profile is also restored at boot.

Waveform save and recall are not available. The saved settings do not include the Domain choice or Reference/Tolerance editor values. Calibration persistence is separate from settings persistence.

Flash storage and interruption

Storage uses one 1 KiB Flash page at 0x0800FC00–0x0800FFFF. The current records use at most 272 bytes: a 32-byte envelope, a 36-byte settings record and up to 204 bytes of calibration data, leaving 752 bytes unused. CRC, readback and commit validation reject incomplete records.
The page is not erased automatically at boot. Explicit SAVE CONFIG or SAVE CAL erases/programs the page. Reset or power loss during that operation can lose the previous saved settings or calibration; wait for SAVED. The older storage layout is not migrated, and its previous data is not preserved.

Flash endurance is finite; avoid unnecessary saves. Storage interruption recovery and endurance have not been fully qualified.

9 Optional DC calibration

Leave the nominal profile in place unless you have suitable, stable reference-voltage equipment. The onboard test output is not a calibration standard. Calibration converts ADC codes to voltage for DC measurements; it does not qualify AC response, bandwidth, the clock or probe attenuation.

The procedure uses two DC references to fit a profile and a separate third DC reference to check it. Each must be a fresh held live record with unchanged physical switch settings, at least 64 samples, spread no greater than 32 ADC codes, and no rail samples. The two fitting records must differ by at least 64 ADC codes. The third must lie at least 16 codes inside both fitted endpoints and pass the selected tolerance. A reference value is entered for each record. Enter values in microvolts (μV): $1{,}000{,}000\,\mu\mathrm{V} = 1\,\mathrm{V}$. These editor values are not input ratings.

To acquire each reference, leave the calibration fields with **SEL**, select an ordinary field such as Timebase, then use **OK** for RUN and HOLD. **OK** inside a calibration field never starts acquisition.

1. Set DC coupling and one fixed range. In HOLD, choose `BEGIN` and execute. Apply reference 1, set its known value in Calibration reference, RUN and acquire, then HOLD and choose `ACCEPT REF`.
2. Apply a different reference 2. Update Calibration reference to its known value, acquire a fresh record, HOLD, then choose `ACCEPT REF` again.
3. Apply independent reference 3. Update Calibration reference to its known value and set Calibration tolerance to the allowed error. Acquire a fresh record, HOLD, and choose `ACCEPT REF`. The session should show `VERIFIED`.
4. Choose `INSTALL` and execute. The profile is now installed in RAM for the matching DC range. To keep it across restart, select `SAVE CAL` in Storage and execute separately.
5. Choose `ABORT` to discard a pending candidate. Abort leaves any already installed profile in place.

In Reference and Tolerance, **OK** cycles the decimal step (powers of ten); it does not move a cursor digit by digit. **+** / **-** changes the numeric value by that step. Replace an unsuitable held record with a fresh one. A switch change, failed two-point fit or failed third-point check requires `BEGIN` and a new session. Installation and saving do not certify hardware accuracy; an external comparison is still required.

10 Troubleshooting

Screen status or symptom	What to do
No trace or an empty plot	Choose AUTO, return to RUN and wait for a record. Set vertical and horizontal position to zero; a panned or zoomed held view can lie outside the captured data.
`WAIT INPUT`, `Switches settling`	Check the physical slide-switch positions and allow them to settle.
`WAIT TRIG`	In Normal or Single, check that the input crosses the selected trigger level and edge.
`FFT NEEDS FROZEN FRAME`	In RUN, wait for a complete capture; check the trigger if it is waiting. A held record also works.
`FFT NEEDS HELD UNIFORM BLOCK`, `FFT NEEDS 512 SAMPLES`	Wait for capture to finish, or use a held record with at least 512 uniformly timed samples.
`FFT TIMING UNKNOWN`	Select a timebase with a valid timer plan, then acquire a fresh record.
`Peak --`	The mean-removed block has no nonzero positive-frequency bin. A DC-only input is removed before analysis.
`-- CYCLES`	Select a wider time span so more periods fit in the fixed record. For the onboard signal, try 1 ms/div.
`RAW: no volts`	Voltage calibration is unavailable. Read ADC codes only or restore/use an appropriate profile.
`ADC CLIP`, `RAW ADC CLIP`	Reduce the input or choose a suitable physical range, then capture again. Clipping can add false harmonics in Spectrum view.
`OLD range/coupling held`	The held record belongs to the previous switch setting. Run a fresh capture.
`HOLD TO SAVE`	Wait in HOLD for a complete, fully processed record, then retry.
`EMPTY`, `READ ONLY`, `DISABLED`	The requested storage item is absent or the action is unavailable; check selection and status.
`ERROR`, `Save failed; kept; OK retry`	The previous record remains held when available. Wait, then retry once with a fresh **OK**; if it repeats, stop using storage and restart normally.
`FAULT`	Wait for the fault state, release all buttons, then use a fresh **OK** retry if offered.

If a button appears ignored, press only one at a time and release it fully before the next action. In Storage or Calibration, remember that the first **OK** may only stop acquisition. Use a second press after release to execute. Do not reset during a save.

11 Experimental USB export

Opt-in bench test

USB export has been tested on the reference board using the explicitly enabled `DSO_USB_STOCK_BENCH_TRIAL` profile (default off). The trial uses the board's fixed D+ pull-up and is not an electrically qualified production configuration. Power the scope from its normal 9 V input; the USB cable carries data. The USB connection joins the instrument and PC grounds.

For the verified held-frame read, make a local capture with the scope controls and place the scope in HOLD. Wait until the record is complete. The MATLAB client reads the existing frame; it does not change scope settings or write Flash settings/calibration.

In MATLAB R2025b, use the COM port shown in Windows Device Manager. The verified reference setup appeared as `COM7`; replace `COMx` with the port assigned by your PC. Run from the repository root:

```
addpath("host/matlab")
client = DsoUsbClient("COMx");
cleanup = onCleanup(@() delete(client));
frame = client.getHeldFrame();
plot(frame.TimeSeconds, double(frame.Raw));
xlabel("Time from trigger (s)");
ylabel("ADC count");
grid on;
```

Each frame contains 12-bit ADC codes, converted volts when available, and a time axis based on the capture timer. The verified record used nominal calibration kind 1. Its 1,024 samples used a configured interval of 2 μs (72 MHz divided by 144). Repeated reads of one held record matched. An earlier stock-board run also passed a four-second slow-host lease check.

The client also exposes `capture` and `captureRepeated`. Put the scope in HOLD before requesting Single. The client waits for the scope to finish its foreground work. In the R6 bench run, MATLAB returned three distinct 1,024-sample records in 3.36 s. A separate protocol test measured about 40–43 kB/s of raw sample payload and verified cancellation and timeout cleanup. Repeated calls produce separate captures with acquisition and presentation gaps. See the MATLAB instructions and USB protocol documentation for setup and limits.

The R6 bench image passed three debugger-controlled restarts with Windows CDC enumeration. After dropping DTR and reopening the serial port, the old lease token was rejected and a fresh download returned the same held record. Software tests cover USB bus suspend and resume handling.

12 Reference and scope

Timebase catalogue (time per horizontal division):

Unit	Values
Microseconds (μs/div)	10, 20, 50, 100, 200, 500
Milliseconds (ms/div)	1, 2, 5, 10, 20, 50, 100, 200, 500
Seconds (s/div)	1, 2, 5

This firmware supports 18 triggered settings from 10 μs/div to 5 s/div. The original DSO138 and other firmware may describe longer settings; they are not available in this firmware. The configured fast acquisition rate reaches 500 ksample/s; this is a timer plan, not an independently measured sample rate, jitter or timebase accuracy. The horizontal scale, display refresh and acquisition rate are different quantities.

Voltage and probe limits. Nominal range labels and `NOM` conversion use board coefficients; they are not a production calibration. `CAL` indicates a measured profile is installed, but does not replace independent verification. `RAW` provides codes, not input volts. No probe attenuation is selected by firmware; multiply displayed input estimates by the probe factor when applicable. Do not treat the onboard signal as an absolute reference. No maximum safe input is specified here: follow the board's own input markings and limits.

Firmware identity and qualification

The deployed R6 bench image uses 64,408 of 64,512 application bytes, leaving 104 bytes. SRAM includes 112 bytes of initialized data, 13,192 bytes of BSS, a 2,048-byte stack reservation and 5,128 unallocated bytes. Heap use is zero. The observed stack write watermark was 928 bytes in the exercised workloads. The static engineering budget is 1,544 bytes. The final 1 KiB Flash page holds up to 272 bytes of settings and DC calibration data.

With the onboard test signal and the configured 500 kS/s rate, the target FFT peak agreed with an independent calculation from the same ADC samples. A held capture retained its raw samples and frozen calibration and settings across Scope and Spectrum transitions. Acquisition continued in both modes during short USB traffic tests. The onboard test signal shares the MCU clock. Absolute timing, analog and spectral accuracy, long-term storage, and electrical USB compliance require separate qualification. The ordinary source profile keeps USB disabled.

Manual revision 2.0, 3 November 2026. Operator reference sources: `firmware/App/Src/dso_ui_model.cpp`, `firmware/App/Src/dso_ui_controller.cpp`, `firmware/App/Src/dso_scaling.cpp`, `firmware/App/Src/dso_spectrum.cpp`, and `firmware/App/Inc/dso_timebase_presets.hpp`.